

# Checkpoint 2 – Infrastructure Configuration

Big Data End-to-End Streaming Pipeline

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## 1. Introduction

This report presents the complete infrastructure of the Big Data pipeline deployed on a cluster of four virtual machines (master, master2, worker1, worker2). The purpose of this checkpoint is to show:

- the complete configuration of the cluster (HDFS, Spark, Kafka),
- the assignment of roles and IPs,
- the global architecture diagram,
- a demonstration showing an active Kafka topic and Spark being able to run jobs.

## 2. Cluster Map – Machines and IPs

| VM      | Main Role                                                        | IP Address |
|---------|------------------------------------------------------------------|------------|
| master  | NameNode, ResourceManager, Spark Master, Kafka Broker, ZooKeeper | 10.0.0.88  |
| worker1 | DataNode, NodeManager, Spark Worker                              | 10.0.0.89  |
| worker2 | DataNode, NodeManager, Spark Worker                              | 10.0.0.92  |
| master2 | High availability (standby NameNode + RM)                        | 10.0.0.93  |

## 3. Component Roles

### HDFS

- **NameNode:** master
- **SecondaryNameNode:** master
- **DataNodes:** worker1, worker2

### YARN

- **ResourceManager:** master
- **NodeManagers:** worker1, worker2

## Spark

- **Spark Master:** master (port 8080)
- **Spark Workers:** worker1, worker2
- **Current state:** 2 active workers, 4 cores, 5.7 GB RAM

## Kafka

- **Kafka Broker:** master (port 9092)
- **ZooKeeper:** master (port 2181)

## Time-Series DB / Grafana

- Not installed for this checkpoint (planned for checkpoint 3).

## 4. Global Architecture

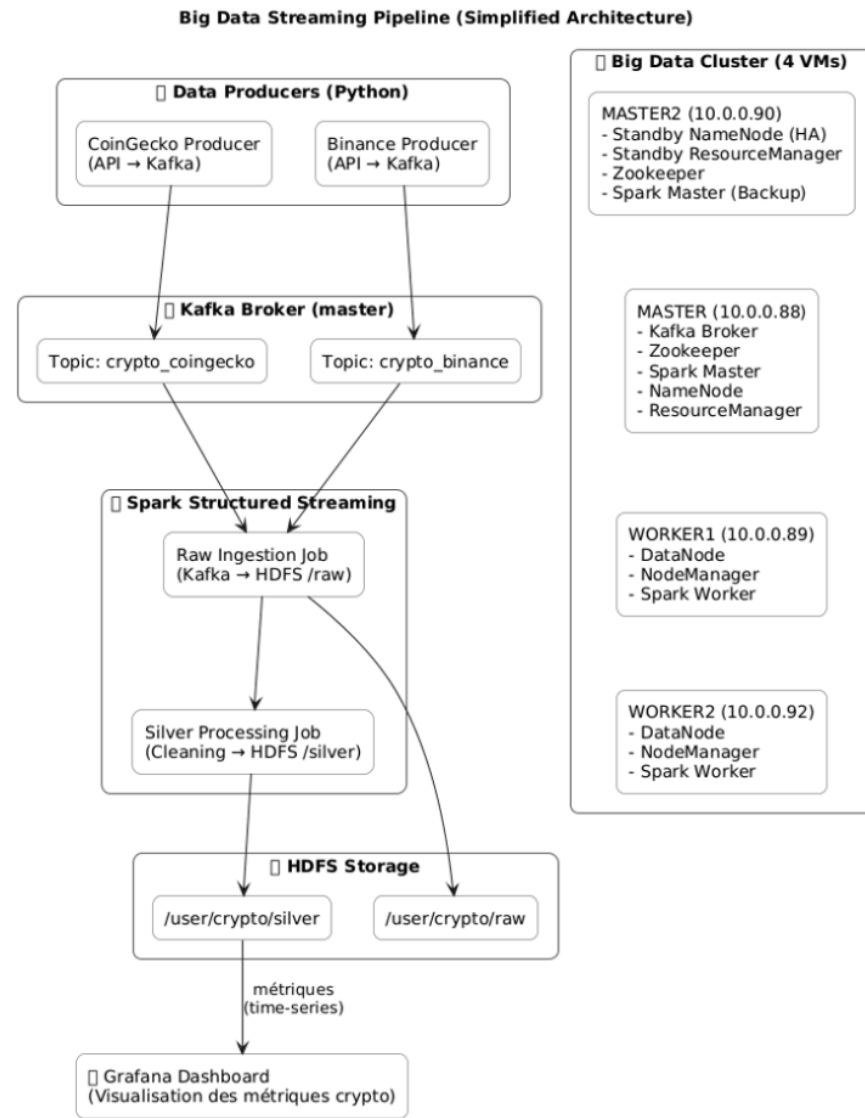


Figure 1: Big Data End-to-End Streaming Pipeline Architecture

## 5. Demonstration (Kafka + Spark)

Here is proof that the cluster works and that messages are correctly ingested.

### 5.1 Kafka – Messages received in the topic

```
(venv_spark) admin@master:~/opt/kafka/kafka_2.13-3.7.0$ bin/kafka-console-consumer.sh --bootstrap-server master:9092 --topic crypto_coingecko --from-beginning
{"bitcoin": {"usd": 90104}, "ethereum": {"usd": 3032.18}}
{"bitcoin": {"usd": 90089}, "ethereum": {"usd": 3030.33}}
{"bitcoin": {"usd": 90089}, "ethereum": {"usd": 3030.33}}
{"bitcoin": {"usd": 90089}, "ethereum": {"usd": 3030.33}}
{"bitcoin": {"usd": 90089}, "ethereum": {"usd": 3030.33}}
{"bitcoin": {"usd": 90089}, "ethereum": {"usd": 3030.33}}
```

Figure 2: Messages received in the Kafka topic

## 5.2 Spark – Connected workers and running application

```
(venv_spark) adm-mcsc@master:~/producers$ python3 coingecko_producer.py
[+] Starting CoinGecko producer...
Message sent: {'bitcoin': {'usd': 90098}, 'ethereum': {'usd': 3031.9}}
Message sent: {'bitcoin': {'usd': 90098}, 'ethereum': {'usd': 3031.9}}
Message sent: {'bitcoin': {'usd': 90098}, 'ethereum': {'usd': 3031.9}}
Message sent: {'bitcoin': {'usd': 90106}, 'ethereum': {'usd': 3032.62}}
Message sent: {'bitcoin': {'usd': 90106}, 'ethereum': {'usd': 3032.62}}
Message sent: {'bitcoin': {'usd': 90106}, 'ethereum': {'usd': 3032.62}}
```

Figure 3: Spark – Connected workers and running application

## 6. Conclusion

The Big Data cluster composed of four machines is now fully operational: Hadoop, YARN, Spark, and Kafka are running correctly.

The Python producers send data to Kafka, and Spark Structured Streaming successfully reads the topics and writes into HDFS.

The system is ready for Checkpoint 3:

- creation of Silver/Gold zones,
- integration of a time-series database (InfluxDB / TimescaleDB),
- Grafana dashboard.